Given these sample points:

[0.89698105 0.72562797 0.17540431 0.70169437]
[0.8893564 0.49958786 0.53926886 0.50878344]
[0.25094624 0.03369313 0.14538002 0.49493242]
[0.34696206 0.0062504 0.76056361 0.61302356]
[0.12487118 0.12977019 0.38440048 0.2870761]
[0.80130271 0.50023109 0.70664456 0.19510284]
[0.24770826 0.06044543 0.04218635 0.44132425]
[0.74670224 0.7570915 0.36935306 0.20656628]
[0.40066503 0.07257425 0.88676825 0.24384229]
[0.6260706 0.58675126 0.43880578 0.77885769]
[0.95713529 0.59764438 0.76611385 0.77620991]
[0.73281243 0.14524998 0.47681272 0.13336573]
[0.65511548 0.07239183 0.68715175 0.08151656]
[0.21973443 0.83203134 0.48286416 0.08256923]
[0.48859419 0.2119651 0.93917791 0.37619173]
[0.16713049 0.87655456 0.21723954 0.95980098]
[0.21691119 0.16608583 0.24137226 0.77006248]
[0.38748784 0.80453226 0.75179548 0.72382744]
[0.98562189 0.66693268 0.15678328 0.8565348]
[0.03782483 0.66485335 0.16198218 0.25392378]
[0.68348638 0.9027701 0.33541983 0.99948256]
[0.17034731 0.75695908 0.27652049 0.5312315]
[0.85965692 0.91959232 0.20613873 0.09779683]
[0.28213837 0.50598691 0.53053084 0.09630162]
[0.32607578 0.4723669 0.453192 0.10588734]
[0.94838936 0.89451301 0.85163782 0.55219629]
[0.66495539 0.04656628 0.11677747 0.79371778]
[0.57776561 0.42877174 0.42582587 0.24900741]
[0.73861301 0.48210263 0.70936644 0.50397001]
[0.8548108 0.49396462 0.73530997 0.80809201]
[0.371549 0.410236 0.317621 0.438427]
[0.611853 0.139494 0.292145 0.366362]
[0.404655 0.429048 0.22677 0.455883]
[0.45607 0.785176 0.199674 0.514234]
[0.592415 0.04645 0.607545 0.170524]
[0.065052 0.948886 0.965632 0.808397]
[0.304614 0.097672 0.684233 0.440152]
[0.122038 0.495177 0.034389 0.90932]

And these results:

-22.108287785435817
-14.601396631953161
-11.699932463834426
-16.053765110568296
-10.069633426012889
-15.487082537755288
-12.68168497681565
-16.02639976898352
-17.049234647535723
-12.741765988316185
-27.316396356142608
-13.527648872277506
-16.679115197479913
-16.50715856428234
-17.81799933882949
-26.561820829779602
-12.758324216781741
-19.441557624451722
-28.90327367383377
-13.702746938161251
-29.427091401989347
-11.565741989583191
-26.857786437773637
-7.9667753510303925
-6.702089254839066
-32.625660215962455
-19.98949792679556
-4.025542281908162
-13.122782331852594
-23.139428397634813
-0.3306618740636291
-7.7047651736267895
-2.6353774299788344
-11.465200220133969
-12.89826399509089
-35.98495615529748
-10.35316016548428
-22.09963492120315

A small positive number is bigger than a large negative number

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially doing **black-box optimization** (likely Bayesian / surrogate-based), and your goal is to **maximize the objective** given past samples.

1. Identify the current best point

Since *less negative = better*, the maximum so far is:

- **-0.3307** at
  x ≈ [0.3715, 0.4102, 0.3176, 0.4384]

Next best:

- **-2.6354** at
  [0.4047, 0.4290, 0.2268, 0.4559]
- **-4.0255** at
  [0.5778, 0.4288, 0.4258, 0.2490]

2. What this tells us

- The optimum is likely in a **region around ~ (0.35–0.45, 0.4–0.45, 0.2–0.35, 0.4–0.5)**
- The function appears **smooth locally** (neighboring points are also relatively good)
- You should **exploit locally**, not explore far away

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3. Where to sample next (Top-k = 3 candidates)

Using a **low temperature (0.1)** → mostly exploit
Top-k = 3 → propose **3 strong candidates near the best region**

🔹 Candidate 1 (best guess – local refinement)

```csharp
[0.36, 0.42, 0.30, 0.46]
```

- Slight perturbation of best point
- Moves toward second-best sample

🔹 Candidate 2 (interpolate best + 2nd best)

```csharp
[0.39, 0.42, 0.27, 0.45]
```

- Blends top two performers
- Likely inside high-performing basin

🔹 Candidate 3 (explore slight variation in 3rd dim)

```csharp
[0.37, 0.41, 0.35, 0.43]
```

- Tests sensitivity in dimension 3 (which varies more)

---

4. Strategy explanation

With your constraints:

- **Temperature = 0.1** → minimal randomness → stay near best
- **Top-k = 3** → focus on top regions only
- **Top-p = 3** (interpreted as tight sampling mass) → reinforces exploitation

So the correct move is:

> **Sample very close to the current best point, with small structured perturbations**

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5. Key insight

Your data strongly suggests:

- Good region is **narrow and localized**
- Random exploration (like your earlier samples) is inefficient now

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✅ Final answer

Sample next at:

1. `[0.36, 0.42, 0.30, 0.46]`
2. `[0.39, 0.42, 0.27, 0.45]`
3. `[0.37, 0.41, 0.35, 0.43]`

If you want, I can fit a quick surrogate model (e.g., Gaussian Process intuition) and give a **single best predicted optimum**, not just candidates.